UNASSIGNED — Ruban $T(2,3)$ Reproduction Falsification Protocol and Final Report

SST-Workbench

30 October 2026 | v0.1.1

1 Identification, scope, and status

Project	Ruban $T(2,3)$ publication reproduction
Version	v0.1.1
Catalog	UNASSIGNED (standalone, pending SST-Workbench admission)
Framework report template	v1.1.0 RC2, SST-FALSIFIER-RUNTIME-2
Framework execution	NOT EXECUTED; template-aligned development only
Scientific result	UNRESOLVED

Report provenance. This document is based on the authentic Framework v1.1.0 RC2 report template; it follows the section-marker/report-insertion contract, but the runtime snapshot, exact `FREEZE` validation and native certification were not installed or invoked. The scientific protocol is **not frozen**. No framework certification is claimed.

2 Research question and objective

Can the regularized Biot–Savart filament model of Ruban (2018), initialized as the analytical trefoil $T(2,3)$ at $B_0 = 0.154$, $\Lambda = 1.4 + \frac{1}{2}\ln 200$ and $\delta/R_0 = 0.05$, achieve operational survival for dimensionless time $t \geq 300$, translating more than $1200R_0$, without loss of numerical validity? This is a *publication reproduction*, not a test of the full three-dimensional Euler equations, volume conservation of a thick tube or internal twist.

3 Competing hypotheses and allowed conclusions

H_0**:** The published long-lived interval is not reproduced by the specified code and refinement schedule with the stated operational rules. H_1**:** A test at $B_0 = 0.154$ reaches $t \geq 300$, $|z(t) - z(0)| \geq 1200$ with acceptable conservation and resolved spatial/temporal error. If the solver violates numerical quality or the independent refinement is absent, the result is **UNRESOLVED**, not evidence against the physical claim. A partial run is not a test of survival beyond its attained time.

4 Assumptions, domain of validity, and known edge cases

A1. Single closed analytical $T(2,3)$ filament in an unbounded fluid, with no externally imposed velocity.

A2. Dimensionless units $R_0 = 1$ and circulation $\Gamma = 2\pi$. Smooth closed periodic curve; one discretized centerline per seed.

A3. Rosenhead-type regularization δ and a separate compensating local-induction coefficient; δ is not identified with the physical core radius.

A4. Deterministic low-wavenumber perturbation with a recorded seed. Different resolutions and B_0 values are *not* independent source providers.

The limits $B_0 \to 0$ (non-embedded multiply traversed circle), $\delta \to 0$ (regularization singularity), and underresolved near-contact geometries require separate analysis. No finite-core Euler or reconnection model is claimed.

5 Symbols, units, and dimensional consistency

Symbol	Meaning	Units here
$\mathbf{X}$	filament centerline	R_0
β	periodic parameter	dimensionless
B_0	torus minor/major ratio	dimensionless
δ	Biot–Savart regularization	R_0
Λ	physical slender-core logarithm	dimensionless
$\widetilde{\Lambda}_0$	$\Lambda + \ln(\delta/R_0)$	dimensionless
t	normalized simulation time	R_0^2/Γ scaling
H	discretized filament Hamiltonian	normalized model energy

With physical circulation units $[\Gamma] = L^2T^{-1}$, the nonlocal line integral carries units L^{-1} and is multiplied by Γ to become velocity. In the chosen Ruban dimensionless convention, the integral coefficient is $1/2$; all computed coordinates and times are dimensionless. The local term has curvature units L^{-1} and the same time-normalization scale.

6 Exact mathematical model and observables

E1 - continuous operator.

$$\partial_t \mathbf{X}(\beta) = \frac{1}{2}\int_0^{2\pi} \frac{\mathbf{X}_\beta(\beta') \times [\mathbf{X}(\beta) - \mathbf{X}(\beta')]}{(|\mathbf{X}(\beta) - \mathbf{X}(\beta')|^2 + \delta^2)^{3/2}}\, d\beta' + \frac{\widetilde{\Lambda}_0}{2}\frac{\mathbf{X}_\beta \times \mathbf{X}_{\beta\beta}}{|\mathbf{X}_\beta|^3}. \tag{1}$$

E2 - discrete update. Fourier spectral derivatives in β, periodic trapezoidal quadrature over all samples (including the finite-regularization diagonal), and classical explicit fourth-order Runge–Kutta. The step is capped by an initial-value-model local-induction spectral bound evaluated on every accepted state:

$$\Delta t_{\text{allowed}} = \min\left(\Delta t_{\text{requested}},\ s\,\frac{4\sqrt{2}}{|\widetilde{\Lambda}_0|}\left(\frac{\min_i |\mathbf{X}_{i+1} - \mathbf{X}_i|}{\pi}\right)^2\right), \qquad s = 0.4. \tag{2}$$

This is a *heuristic conservative stability cap*, not a rigorous stability bound for the nonlocal coupled operator. The code also clips to upcoming diagnostic/output times; underflow and geometric degeneration stop the run.

E3 - discrete Hamiltonian and diagnostics. The code uses the discretized Hamiltonian corresponding to the regularized interaction and local-induction term:

$$H = \frac{\pi}{2}\int_0^{2\pi}\int_0^{2\pi} \frac{\mathbf{X}_\beta(\beta)\cdot\mathbf{X}_\beta(\beta')}{\sqrt{|\mathbf{X}(\beta) - \mathbf{X}(\beta')|^2 + \delta^2}}\, d\beta'\, d\beta + \pi\widetilde{\Lambda}_0 \int_0^{2\pi} |\mathbf{X}_\beta|\, d\beta. \tag{3}$$

The double and single integrals are evaluated with periodic trapezoidal quadrature, using the same spectral derivatives as in E1.

$$\epsilon_H(t) = \frac{|H(t) - H(0)|}{\max(|H(0)|, 10^{-12})}, \quad D(t) = \mathrm{RMS}_{\text{rigid,cyclic}}(\mathbf{X}(t), \mathbf{X}(0)). \tag{4}$$

E4 - decision thresholds. Require $\epsilon_H \leq 10^{-3}$, segment ratio $\max(\Delta s)/\min(\Delta s) \leq 10$, operational shape RMS $D \leq 0.25R_0$, $t \geq 300$, $|\Delta z| \geq 1200R_0$; thresholds are *local operational choices*, not Ruban's unpublished termination rule. Diagnostics run at intervals no longer than $\Delta t_{\text{diag}} = 0.025$. Scientific qualification still requires convergence.

Machine-readable equation register

- **E1**: `ruban_falsifier/model.py:RubanModel.velocity`, gate G2.
- **E2**: `ruban_falsifier/model.py,runner.py`, gate G2.
- **E3**: `ruban_falsifier/runner.py`, gate G2.
- **E4**: `ruban_falsifier/runner.py`, gate G3.

Framework-runtime validation: NOT EXECUTED.

7 Exact algorithm: equation-to-code mapping

S1. Verify bundled input hashes and analytic seed parameters ($T(2,3)$, B_0).

S2. Compute a ring symmetry benchmark with the same regularized filament operator.

S3. Generate a torustrefoil at N points, optionally with seeded smooth perturbation.

S4. Evaluate all terms in Eq. (1), impose the step ceiling Eq. (2) and integrate by Runge–Kutta.

S5. Check geometry at every accepted step; energy and deformations at every diagnostic checkpoint.

S6. Stop on the first failing numerical threshold, save a complete trace and closed-curve snapshots.

S7. Evaluate independent spatial, temporal and regularization scans at fixed physical end time in a distinct campaign.

S8. Record all gate results without promoting a short trial to the published long-time claim.

Python/Numba float64 is a development reference only. No native C++ certification is present. Artificial damping, smoothing and remeshing are absent.

8 Source, provenance, and independence contract

Ruban (2018), *Long-Lived Quantum Vortex Knots*, supplies the equations and target conditions. The local analytic $T(2,3)$ seed is one generated source family, not an independent provider. The user-supplied `3_1.xyz` is hashed and retained for comparison but is not input to the time integrator. E010 v0.4.0 is geometry-only context. Numerical refinements of the same analytic seed are not independent replications. Exact input/source SHA-256 values accompany each run.

9 Gate DAG and decision semantics

Gate	Meaning
G0	Provenance and input manifest
G1	Circular-ring field symmetry
G2	Numerical validity, energy, segment and step controls
G3	Time > 300 and travel $> 1200R_0$ only with prerequisite numerical quality
G4	Local B_0 stability window: withheld until proper lifetime comparison
G5	Independent temporal/spatial convergence at equal final time
G6	Publication-level numerical reproduction independent audit
G7	Full finite-core Euler and internal core twist: not tested

Status categories: PASS, FAIL, UNRESOLVED, NOT_TESTED. The local runtime does not claim the official v1.1.0 gate DAG, HMAC blindness, reveal phase or certification. For a numerically rejected run, the command exits nonzero even when source and ring gates PASS.

10 Numerical certification, convergence, and error budget

A convergence gate requires matching physical final time at $N = 128, 256, 512$, with consistent refinement $\Delta t \propto N^{-2}$ and smaller time steps; compare centerline distances modulo rigid motion, energy, segment ratio, and velocity. The completed v0.1.1 pilots do not automatically close this gate. Lowering δ changes the modeled regularization and is *not* interchangeable with a fixed-model spatial convergence study. RK4 stability of the local-induction term alone cannot guarantee nonlocal full-operator stability.

11 Backend authority and parity

The current code uses Python, NumPy and Numba with float64 arithmetic on a CPU. There is no verified native C++/OpenMP or SYCL implementation, parity certificate, device manifest, or GPU support. Therefore native and GPU certification gates remain **NOT EXECUTED**; numerical PASS in the current implementation does not imply framework certification.

12 Blindness, opaque identifiers, and reveal

This is an explicitly unblinded attempt to reproduce already-published numerical results. No concealed target labels or private reveal keys are present. The framework reveal phase is disabled; there is no claim of preregistration or independent blinded confirmation.

13 Statistical/null model and multiplicity

Deterministic single-seed parameter sweeps do not form independent randomized trials. Only one analytic seed family is tested; no inferential p-values or confidence intervals are claimed. A true perturbation stability distribution requires preregistered multiple seeds, fixed disturbance amplitudes, and withheld confirmation.

14 Results and automatic run provenance

Executed run

Mode	FULL
Recorded UTC	20261030T080553305422Z
Configuration SHA-256	`9f20118a1041f9afd210cc6076982f9b` (prefix)
Backend	Python/Numba float64 - DEVELOPMENT ONLY

Numerical results

B_0	N	t	ϵ_H	Δz	stop
0.140	256	3.875	1.06e-03	15.012	`ENERGY_DRIFT_STOP`
0.148	256	19.35	1.01e-03	74.167	`ENERGY_DRIFT_STOP`
0.152	256	78.13	1.22e-03	298.118	`ENERGY_DRIFT_STOP`
0.154	256	80.83	1.00e-03	307.462	`ENERGY_DRIFT_STOP`
0.156	256	67.25	1.06e-03	255.517	`ENERGY_DRIFT_STOP`
0.160	256	25.42	1.02e-03	95.881	`ENERGY_DRIFT_STOP`
0.168	256	5.125	1.15e-03	19.115	`ENERGY_DRIFT_STOP`

Gate ledger

Gate	Status
`G0_PROVENANCE`	PASS
`G1_RING_SYMMETRY`	PASS
`G2_NUMERIC_ENERGY`	FAIL
`G3_RUBAN_T300_AND_TRAVEL1200`	UNRESOLVED
`G3_PUBLICATION_ENERGY_7DIGITS`	UNRESOLVED
`G4_STABILITY_WINDOW`	UNRESOLVED
`G5_CONVERGENCE`	UNRESOLVED
`G6_PAPER_REPRODUCTION`	UNRESOLVED
`G7_EULER_FINITE_CORE`	NOT_TESTED
`FRAMEWORK_RUNTIME_CERTIFICATION`	NOT_EXECUTED

Reports of apparent persistence at short times or without G5 convergence are NOT long-time stability certification.

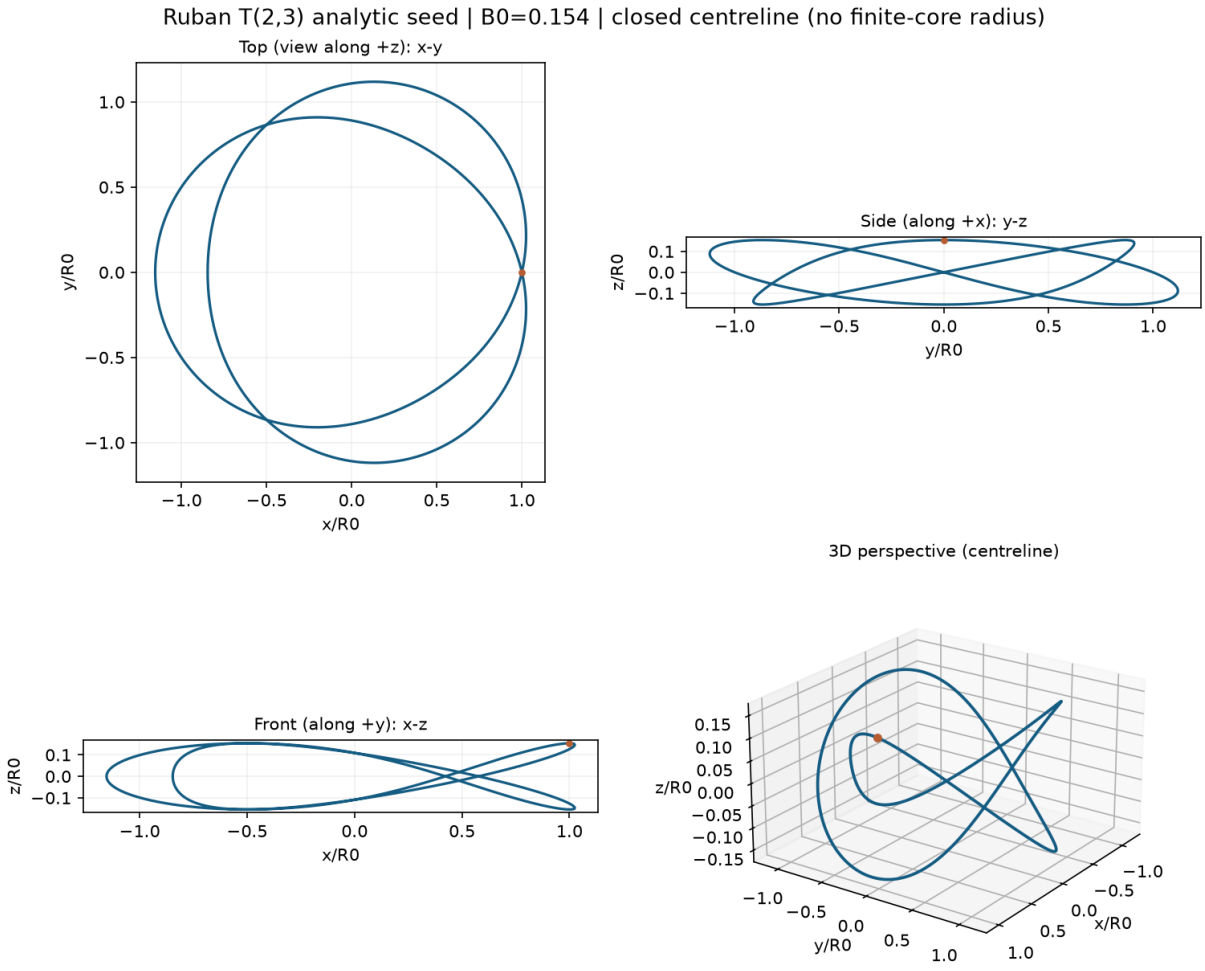


Figure 1: The analytical $T(2,3)$ seed, shown in the requested 2x2 array: top, side, front, and 3D perspective. The plotted line is the *closed centerline only*; its line thickness is not a physical tube radius, and δ is a numerical regularization scale.

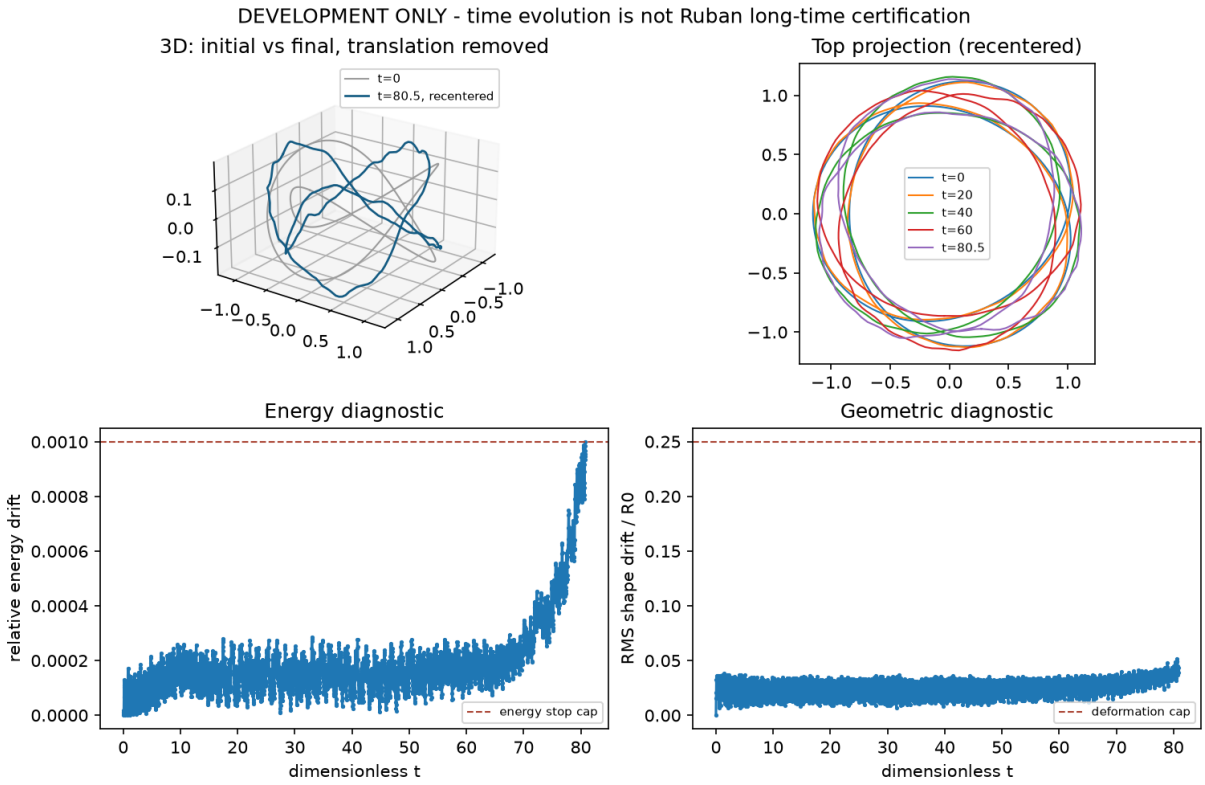


Figure 2: Development-time evolution and diagnostics, referenced to the actual run. Short-time geometric persistence does not establish long-time stability.

15 Interpretation, limits, falsifiable predictions, and minimal follow-up experiments

A short run that reaches its requested time with adequate energy conservation demonstrates only *local numerical viability*. A long run that violates the energy gate is a numerical failure, not a physical refutation of the Ruban configuration. A positive long-time survival remains provisional until stable under space/time refinement. Minimum follow-up: canonical $B_0 = 0.154$ at equal $t = 1$ for $N = 128, 256, 512$, each with halved time steps; then verified $t = 10$, $t = 30$, and ultimately $t > 300$. Internal twist and a dense background filament lattice are excluded from this version to isolate the published mechanism.

15.1 Post-run discussion

No post-run reinterpretation is silently inserted.

16 Reproducibility and exact commands

```
py -m pip install -r requirements.txt
run_all.cmd TEST
run_all.cmd SMOKE
run_all.cmd BASIC
run_all.cmd CONVERGE
run_all.cmd FULL
```

`run_all.cmd FREEZE`, `CERTIFY`, `REVEAL` are **refused** until an authenticated Framework v1.1.0 runtime instance has been installed and validated. Outputs are written to a version-specific directory and published in sibling ZIP/SHA-256 and PDF files. Existing output roots are not overwritten.

17 References

References

[1] V. P. Ruban (2018), “Long-Lived Quantum Vortex Knots,” *JETP Letters* 107, 307–310, DOI: https://doi.org/10.1134/S0021364018050120.

[2] V. P. Ruban (2018), “Quasi-Stable Configurations of Torus Vortex Knots and Links,” *Journal of Experimental and Theoretical Physics* 127, 581–586, DOI: https://doi.org/10.1134/S106377611809008X.

[3] P. G. Saffman (1992), *Vortex Dynamics*, Cambridge University Press, DOI: https://doi.org/10.1017/CBO9780511624063.

[4] B. Gilbert (2016), *Database of Ideal Knots 3–10 Crossings*, numerical dataset, entry 3:1:1, available from user-supplied library record.